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The central task of creating the selective water purification membrane has two sub-tasks: (1) creation of a supramolecular phospholipid bilayer scaffold that is robustly anchored to a porous substrate and (2) development of a cell-free process for the biosynthesis of aquaporin coupled with its insertion into the lipid bilayer. Each of these sub-tasks has several elements, as described in the following.
Sub-task 1: Anchor a Phospholipid Bilayer Scaffold to a Porous Substrate

This sub-task has three elements that may be organized spatially beginning with the substrate and progressing to the final functional membrane: (i) selection or fabrication of a porous substrate that will provide mechanical support for the lipid bilayer and permit transport and collection of the purified water, (ii) development of a scheme for robustly tethering the lower leaflet of the lipid bilayer, (iii) development of the process protocol whereby the final bilayer is assembled with functional aquaporin.

Task Description:

(i) Porous substrate: Two classes of porous substrates are being studied. The first consists of commercially available materials e.g. ultrafiltration membranes based on regenerated cellulose or anodized aluminum (Anodisc). The second consists of aerogel or xerogel substrates.

(ii) Tethering schemes: Much of the initial development of synthetic biomembranes began with “supported” lipid bilayers deposited on quartz or glass and separated from the substrate by a thin (10-20 Å) water layer. However, transmembrane proteins that are incorporated into these supported lipid bilayers have the potential to lose their functionality through non-physiological interactions with the substrate. This has led to extensive efforts to separate the bilayer from the solid support. One of our assembly strategies uses an intervening hydrophilic polymer cushion that is covalently bound to the substrate as well as firmly anchored to the lipid bilayer. This method is advantageous because it does not depend on electrostatic interactions, such that ionic strength and pH can be changed depending on the specific requirements of the incorporated proteins. Another approach to forming anchored lipid membranes uses biological interactions. Streptavidin is a roughly cylindrical protein that binds very tightly to biotin at both ends of the cylinder. Thus, it potentially can serve as an anchor to immobilize lipid bilayers containing biotinylated lipids.

(iii) Bilayer assembly: Lipid bilayer vesicles, or liposomes, have been widely studied as delivery vehicles for drugs, genetic material, and other biomolecules into living cells. Understanding how the molecular processing protocol influences the structural and functional characteristics of these supramolecular assemblies is critical to their application in the tethered bilayers of the water purification membrane. Specifically, we are interested in the process of vesicle fusion whereby vesicles, which may contain aquaporin, interact with a substrate in a controlled manner such that the vesicles rupture and form a defect-free planar bilayer. Two key elements are that we wish to understand the state of organization of the vesicles in the initial suspension and we wish to follow the kinetics of the adsorption on the substrate. The first of these is accomplished with dynamic light scattering, which is being used to follow the evolution of structure in relatively narrow size distribution vesicles prepared by extrusion. The second is addressed with the quartz crystal microbalance, where the frequency shift and dissipation change permit differentiation of adsorbed, intact vesicles from fully formed bilayers.

Sub-task 2: Synthesize Aquaporin via Cell-free Protocol and Insert in Bilayer

In order to achieve the ultimate goal of regeneration of active membranes during space flight, we are developing a cell-free biosynthetic approach. This involves the development of reagents that will allow us to remove the E coli signal recognition particle protein and the secretion helper proteins from the cell extract used for protein synthesis. These have become available and we have demonstrated aquaporin expression followed by functional insertion of the aquaporin transmembrane protein. Membrane function in liposome structures is being assessed by water permeability and urea flux measurements.

(i) Expression of functional aquaporin water channels: Membrane proteins have been notoriously difficult to produce, so it is important to demonstrate that cell-free protein synthesis is capable of synthesizing these complex proteins. As our development model, we chose the aquaporin from the bacterium, Escherichia coli, since our cell-free system is based on E. coli cell extracts. In initial experiments to test efficiency, we compared the expression of E.coli’s aquaporin Z to that of our model protein, chloramphenicol acetyl transferase (CAT), a protein that is normally synthesized as a soluble protein inside of E.coli. One common problem with such expression is that much of the produced protein does not fold properly. Instead, it aggregates and forms a precipitate. In fact, we discovered that the total expression of aquaporin Z in our system was very good, and a small, but significant portion remains in solution. This demonstrates that there is a small quantity of membrane vesicles in our cell extract, but this amount is almost certainly not sufficient to allow these high protein yields to be inserted into membranes. When synthetic lipid vesicles are formed from dioleoylphosphatidylcholine (DOPC) and added to the cell-free reaction, there is a significant and dose-dependent increase in the fraction of newly synthesized aquaporin Z that remains soluble. This also occurs with the addition of vesicles composed of extracted E.coli lipids.

(ii) Biotinylation and purification of vesicles: Two isolation methods have been developed to recover natural vesicles after the cell-free synthesis and to measure how much of the new aquaporin was associated with the vesicles. The first uses vesicles that have 2% of the lipids chemically coupled to the vitamin, biotin. Avidin is a modified protein that binds to biotin and this is used to capture the new vesicles that are displaying the biotin. A commercial chromatography resin is available with immobilized avidin, and we have used such columns to purify lipid vesicles. Typically, the vesicles applied to the column are 50 nanometers in diameter, have 2% of their lipid molecules connected to biotin, and also contain radioactive lipids (3H) to facilitate measurement. The second method simply uses dialysis with a 300,000 molecular weight cutoff separation membrane. Smaller molecules such as metabolites and proteins diffuse out of the dialysis chamber leaving the large lipid vesicles behind.

To better understand the performance of the synthesis and purification procedures, we have also developed a method to measure how much lipid is associated with the vesicle fraction. This uses a solvent extraction to quantitatively remove the lipids, a hydrolysis reaction to cleave the phosphate from the lipids, and a standard assay for phosphate concentration to finally estimate the lipid concentration. Since we routinely label our synthesized protein using radioactive leucine, we can accurately calculate the aquaporin Z protein-to-lipid ratio in the vesicles recovered after protein synthesis. The molar ratio for the DOPC synthetic lipid vesicles is routinely 2-3 x 10-3. If the aquaporin Z proteins are properly inserted and folded, this number suggests a vesicle surface coverage of aquaporin Z protein molecules of about 4%. With purified E.coli lipids, the number is about three times as high, suggesting an areal coverage of 12%. Our objective is to mimic biological processes in order to produce enabling technology for highly selective water purification filters and highly sensitive and selective chemical sensors. In both cases, lipid bilayer membranes provide the separation barrier and integral membrane proteins contribute biological function as separators or sensors. Our main separations objective is based on aquaporin, a membrane protein that has evolved to serve as a highly selective separation device that passes only water. Most importantly, it efficiently excludes urea, the major solute in urine and a significant separations challenge. Thus, a lipid bilayer containing functional aquaporin proteins will serve as a light-weight water recovery device for extended space travel. More broadly, this technology offers exciting potential for economical, large-scale water purification here on earth. This interdisciplinary research project comprises a fundamental study of aspects of macromolecular biotechnology in pursuit of an enabling technology required for future human space missions. Just as bacteria are able to regenerate failed proteins, a significant advantage of this technology is the opportunity to replace failed filters and sensors during extended space travel. For example, a set of anchored lipid bilayer/filter assemblies could be carried on board in addition to the reagents required to conduct cell-free protein synthesis. By carrying small quantities of a variety of DNA templates, astronauts would be able to synthesize any of the filters or sensors that had failed. The molecular process first will reproduce the biological process of transcription to produce the messenger RNA

	based on the DNA template added. Then the process of translation uses the ribosomes in bacterial cell extracts to polymerize amino acids into the desired polypeptide. As the polypeptide is synthesized, it will be extruded out of the ribosome and into the anchored membrane in order to replicate the natural process of membrane protein insertion and folding to produce a functional filter or sensor. We believe that our aquaporin biomembrane filter will offer a simpler and less energy-intensive approach than current methods for water recovery from urea, which include reverse osmosis or distillation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of a robust device for water purification has immense applications in areas outside of manned space exploration. This is an exciting area for future work.
Task Progress:	No progress report this period.
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